

[BUG]lightning.gpu wrong QAOA results

<https://github.com/PennyLaneAI/pennylane/issues/2745>

ROW 13

[BUG]lightning.gpu wrong QAOA results #2745

Closed



duanyh12 opened on Jun 19, 2022

...

Expected behavior

For a QAOA max cut problem, here I am comparing the qnode results from 4 different devices:

1. lightning.qubit shots=None
2. lightning.gpu shots=None
3. lightning.gpu shots=1
4. lightning.gpu shots=8192 (some large enough value)

Code goes:

```
import os
import pennylane as qml
from pennylane import numpy as np
import networkx as nx
from pennylane import qaoa

n_qubits = 10
edges = 20
seed = 197
params_init = np.ones((2, 2))
dev1 = qml.device("lightning.qubit", wires=n_qubits, shots=None)
dev2 = qml.device("lightning.gpu", wires=n_qubits, shots=None)
dev3 = qml.device("lightning.gpu", wires=n_qubits, shots=1)
dev4 = qml.device("lightning.gpu", wires=n_qubits, shots=8192)

g = nx.gnm_random_graph(n_qubits, edges, seed=seed)
cost_h, mixer_h = qaoa.maxcut(g)

def qaoa_layer(gamma, alpha):
    qaoa.cost_layer(gamma, cost_h)
    qaoa.mixer_layer(alpha, mixer_h)

def qaoa_circuit(params, n_qubits, depth):
    for w in range(n_qubits):
        qml.Hadamard(wires=w)
    gammas = params[0]
    alphas = params[1]
    qml.layer(qaoa_layer, depth, gammas, alphas)

@qml.qnode(dev1)
def cost1(params):
    qaoa_circuit(params, n_qubits, 2)
    return qml.expval(cost_h)

@qml.qnode(dev2)
def cost2(params):
    qaoa_circuit(params, n_qubits, 2)
    return qml.expval(cost_h)

@qml.qnode(dev3)
def cost3(params):
    qaoa_circuit(params, n_qubits, 2)
    return qml.expval(cost_h)

@qml.qnode(dev4)
def cost4(params):
    qaoa_circuit(params, n_qubits, 2)
    return qml.expval(cost_h)

print(params_init)
print(dev1.name)
print(cost1(params_init))
print(dev2.name)
print(cost2(params_init))
print(dev3.name)
print(cost3(params_init))
print(dev4.name)
print(cost4(params_init))
```

I would expect `cost1(params_init)` to match `cost2(params_init)` and to roughly match `cost4(params_init)`. `cost3(params_init)` should be an integer. Let me know if I am mistaken.

Actual behavior

```
| 2022-06-18T19:42:07.895-07:00 | [[1. 1.] [1. 1.]]
| 2022-06-18T19:42:07.895-07:00 | Lightning Qubit PennyLane plugin
| 2022-06-18T19:42:07.895-07:00 | -10.24677464628668
| 2022-06-18T19:42:07.895-07:00 | PennyLane plugin for GPU-backed Lightning device using NVIDIA cuQuantum SP
| 2022-06-18T19:42:07.895-07:00 | -0.0018589484246231427
| 2022-06-18T19:42:07.895-07:00 | PennyLane plugin for GPU-backed Lightning device using NVIDIA cuQuantum SP
| 2022-06-18T19:42:07.895-07:00 | -0.0018589484246231427
| 2022-06-18T19:42:07.895-07:00 | PennyLane plugin for GPU-backed Lightning device using NVIDIA cuQuantum SP
| 2022-06-18T19:42:07.895-07:00 | -0.0018589484246231427
```

Additional information

No response

Source code

No response

Tracebacks

No response

System information

```
2022-06-18T19:42:09.896-07:00 Name: PennyLane
2022-06-18T19:42:09.896-07:00 Version: 0.25.0.dev0
2022-06-18T19:42:09.896-07:00 Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
2022-06-18T19:42:09.896-07:00 Home-page: https://github.com/XanaduAI/pennylane
2022-06-18T19:42:09.896-07:00 Author:
2022-06-18T19:42:09.896-07:00 Author-email:
2022-06-18T19:42:09.896-07:00 License: Apache License 2.0
2022-06-18T19:42:09.896-07:00 Location: /opt/conda/lib/python3.8/site-packages
2022-06-18T19:42:09.896-07:00 Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-1
2022-06-18T19:42:09.896-07:00 Required-by: amazon-braket-pennylane-plugin, PennyLane-Lightning, PennyLane-Li
2022-06-18T19:42:09.897-07:00 Platform info: Linux-4.14.275-142.503.amzn1.x86_64-x86_64-with-glibc2.17
2022-06-18T19:42:09.897-07:00 Python version: 3.8.10
2022-06-18T19:42:09.897-07:00 Numpy version: 1.21.4
2022-06-18T19:42:09.897-07:00 Scipy version: 1.7.3
2022-06-18T19:42:09.897-07:00 Installed devices:
2022-06-18T19:42:09.897-07:00 - braket.aws.qubit (amazon-braket-pennylane-plugin-1.6.7)
2022-06-18T19:42:09.897-07:00 - braket.local.qubit (amazon-braket-pennylane-plugin-1.6.7)
2022-06-18T19:42:09.897-07:00 - default.gaussian (PennyLane-0.25.0.dev0)
2022-06-18T19:42:09.897-07:00 - default.mixed (PennyLane-0.25.0.dev0)
2022-06-18T19:42:09.897-07:00 - default.qubit (PennyLane-0.25.0.dev0)
2022-06-18T19:42:09.897-07:00 - default.qubit.autograd (PennyLane-0.25.0.dev0)
2022-06-18T19:42:09.897-07:00 - default.qubit.jax (PennyLane-0.25.0.dev0)
```

```
2022-06-18T19:42:09.897-07:00 - default.qubit.autograd (PennyLane-0.25.0.dev0)
2022-06-18T19:42:09.897-07:00 - default.qubit.jax (PennyLane-0.25.0.dev0)
2022-06-18T19:42:09.897-07:00 - default.qubit.tf (PennyLane-0.25.0.dev0)
2022-06-18T19:42:09.897-07:00 - default.qubit.torch (PennyLane-0.25.0.dev0)
2022-06-18T19:42:09.897-07:00 - lightning.gpu (PennyLane-Lightning-GPU-0.23.0)
2022-06-18T19:42:09.897-07:00 - lightning.qubit (PennyLane-Lightning-0.23.0)
2022-06-18T19:42:09.897-07:00 None
```

Existing GitHub issues

☒ I have searched existing GitHub issues to make sure the issue does not already exist.



duanyh12 added **bug** on Jun 19, 2022



mlxd on Jun 20, 2022 · edited by mlxd

Edits ▾ Member ...

Hi @duanyh12

This is due to the lack of full support for MultiRZ in LightningGPU, as well as the limited shots>0 support in the 0.23 release. If you install the pre-release 0.24 version (soon to be the fully released 0.24 version), this is working as expected for the exact case.

As for the variation expected with the finite-shots>0, let me check.

```
[[1. 1.]
 [1. 1.]]
Lightning Qubit PennyLane plugin
-10.24677464628668
PennyLane plugin for GPU-backed Lightning device using NVIDIA cuQuantum SDK
-10.246774646286685
PennyLane plugin for GPU-backed Lightning device using NVIDIA cuQuantum SDK
-10.246774646286685
PennyLane plugin for GPU-backed Lightning device using NVIDIA cuQuantum SDK
-10.246774646286685
```



mlxd on Jun 20, 2022

Member ...

This required a quick fix. v0.24 will have this resolved, and will be released tomorrow.



duanyh12 mentioned this on Jun 20, 2022

🔗 [Upgrade container pennylane lightning.gpu version to v0.24](#) [amazon-braket/amazon-braket-containers#50](#)



duanyh12 on Jun 21, 2022

Contributor Author ...

Confirming the issue is resolved with v0.24. Really appreciate the help!



duanyh12 closed this as **completed** on Jun 21, 2022